

Introduction to Mathematics and Optimization

January 2024

The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (6.12.2023) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function

$$f(x, y) = x^2 - y$$

and $C \subseteq \mathbb{R}^2$ the subset

$$C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1, (x+1)^2 + (y-1)^2 \leq 1\}.$$

We consider the optimization problem

$$\begin{aligned} \min & f(x, y) \\ & (x, y) \in C \end{aligned} \tag{1}$$

- (a) Prove that f is a convex function. Is f a strictly convex function?

ANSWER:

The Hessian matrix of f is

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

This is a diagonal matrix with non-negative entries and hence positive semidefinite according to Exercise 8.26. Therefore f is a convex function according to Theorem 8.21. f cannot be strictly convex, since $f(0, y) = -y$ is linear. For example, $f((1-t)(0, 0) + t(0, y)) = (1-t)f(0, 0) + tf(0, y)$ for all $t \in \mathbb{R}$. Hence f is not strictly convex according to Definition 6.1.

- (b) Show that C is a subset of $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ and that C is a bounded subset of $\mathbb{R}^2$.

ANSWER:

C is defined as the intersection of $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ and $\{(x, y) \in \mathbb{R}^2 \mid (x+1)^2 + (y-1)^2 \leq 1\}$. Hence C is a subset of $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$. Therefore C is a bounded subset of $\mathbb{R}^2$, since $(x, y) \in C \implies x^2 + y^2 \leq 1$, which is the definition of C being bounded according to Equation (5.16) with $M = 1$.

- (c) Show that C is a compact and convex subset of $\mathbb{R}^2$.

ANSWER:

The functions $f(x, y) = x^2 + y^2 - 1$ and $g(x, y) = (x+1)^2 + (y-1)^2 - 1$ are each continuous, since they are built from continuous functions via $\times$ and $+$. They are convex functions according to Theorem 8.21, since they both have the positive definite diagonal matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

as Hessian matrix. Since

$$C = f^{-1}((-\infty, 0]) \cap g^{-1}((-\infty, 0])$$

it follows from Lemma 4.26 and Exercise 4.22 that C is convex, and from Proposition 5.50, Proposition 5.73, and Proposition 5.51 that C is closed. Since C is bounded according to the previous part, C is compact according to Definition 5.76. Hence C is a compact and convex subset of $\mathbb{R}^2$.

- (d) Which result in the notes ensures that the optimization problem (1) can be solved, and why?

ANSWER:

The function f is continuous as described earlier and C is compact. That the optimization problem can be solved is concluded from Theorem 5.77.

- (e) Show from the KKT conditions that $(-1, 0)$ is not an optimal solution to (1).

ANSWER:

According to Equation (9.24), the KKT conditions for (1) are

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ x^2 + y^2 - 1 &\leq 0 \\ (x+1)^2 + (y-1)^2 - 1 &\leq 0 \\ \lambda_1(x^2 + y^2 - 1) &\leq 0 \\ \lambda_2((x+1)^2 + (y-1)^2 - 1) &= 0 \\ 2x + 2\lambda_1x + 2\lambda_2x + 2\lambda_2 &= 0 \\ -1 + 2\lambda_1y + 2\lambda_2y - 2\lambda_2 &= 0.\end{aligned}$$

Let us assume that $(-1, 0)$ is an optimal solution to (1). Then we know, since the optimization problem is strictly feasible (e.g., with the point $(-1/2, 1/2)$), by Theorem 9.30(i) that the KKT conditions must have a solution with $x = -1$ and $y = 0$. But if we substitute $y = 0$ into the KKT conditions, we get at the end

$$-1 - 2\lambda_2 = 0,$$

which contradicts $\lambda_2 \geq 0$.

- (f) How do we know that the KKT conditions corresponding to (1) actually have a solution?

ANSWER:

Since we know that there exists an optimal solution to (1) by Theorem 5.77, Theorem 9.30(i) shows that the KKT conditions actually have a solution, since the optimization problem is strictly feasible.

PROBLEM 2

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + z^2 - xy + yz - xz$$

and let $C \subseteq \mathbb{R}^3$ be the subset given by

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid x \geq 0, y \geq 1, z \geq 0\}.$$

(a) Show that f is a strictly convex function.

ANSWER:

The Hessian matrix of f is

$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}.$$

according to Definition 8.2. We transform the Hessian matrix to a diagonal matrix via the operations in Section 8.6. First, we add $1/2$ times row 1 to rows 2 and 3:

$$\begin{pmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 1/2 & 3/2 \end{pmatrix}.$$

The corresponding column operations are to add $1/2$ times column 1 to columns 2 and 3. This gives

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 1/2 \\ 0 & 1/2 & 3/2 \end{pmatrix}.$$

Now we add $-1/3$ times row 2 to row 3 and $-1/3$ times column 2 to column 3:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 4/3 \end{pmatrix}.$$

This matrix is a diagonal matrix with positive entries and is hence positive definite according to Exercise 8.26. According to Proposition 3.41, the Hessian matrix is therefore positive definite, and f is hence a strictly convex function according to Theorem 8.21.

(b) Show that C is not a bounded subset of $\mathbb{R}^3$.

ANSWER:

C is not a bounded subset of $\mathbb{R}^3$, since $v = (x, x, x) \in C$ for all $x \geq 1$ and

$$|v|^2 = x^2 + x^2 + x^2 = 3x^2.$$

For a given $M > 0$, there exists $x \geq 1$ such that $|v|^2 = 3x^2 > M$. According to Equation (5.16), C therefore cannot be bounded.

(c) Write down the KKT conditions for the optimization problem

$$\begin{aligned} \min f(x, y, z) \\ (x, y, z) \in C, \end{aligned} \tag{2}$$

and show that $(\frac{1}{2}, 1, 0)$ is an optimal solution to (2) and that the KKT conditions have a unique solution.

ANSWER:

According to Equation (9.24), the KKT conditions for (2) are

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ \lambda_3 &\geq 0 \\ -x &\leq 0 \\ 1 - y &\leq 0 \\ -z &\leq 0 \\ -\lambda_1 x &= 0 \\ \lambda_2(1 - y) &= 0 \\ -\lambda_3 z &= 0 \\ 2x - y - z - \lambda_1 &= 0 \\ 2y - x + z - \lambda_2 &= 0 \\ 2z + y - x - \lambda_3 &= 0.\end{aligned}$$

We substitute $x = 1/2, y = 1, z = 0$ into the last three equations:

$$\begin{aligned}2\left(\frac{1}{2}\right) - 1 - 0 - \lambda_1 &= 0, \\ 2(1) - \frac{1}{2} + 0 - \lambda_2 &= 0, \\ 2(0) + 1 - \frac{1}{2} - \lambda_3 &= 0.\end{aligned}$$

We end up with

$$\begin{aligned}\lambda_1 &= 0 \\ \lambda_2 &= \frac{3}{2} \\ \lambda_3 &= \frac{1}{2}\end{aligned}$$

and all KKT conditions are satisfied. This solution to the KKT conditions is unique by Theorem 9.30(ii), since there is only one solution to the equations

$$\begin{aligned}2x - y - z - \lambda_1 &= 0 \\ 2y - x + z - \lambda_2 &= 0 \\ 2z + y - x - \lambda_3 &= 0.\end{aligned}$$

given x, y, z . Here we use that an optimal solution to (2) is unique by Theorem 6.12.

(d) Is $f(x, y, z) \geq 0$ for all $x, y, z \in \mathbb{R}$?

ANSWER:

We can write

$$f(x, y, z) = \begin{pmatrix} x & y & z \end{pmatrix} A \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

with

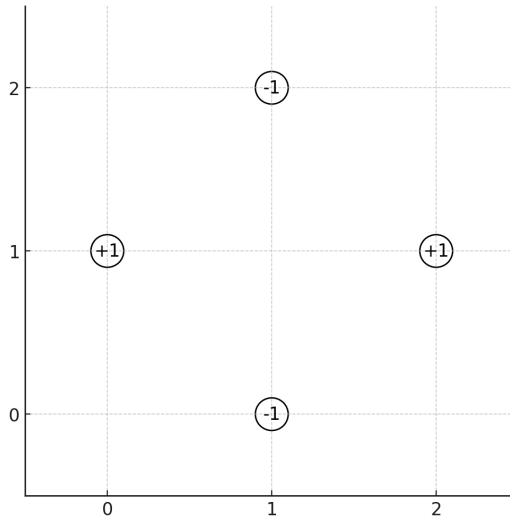
$$A = \begin{pmatrix} 1 & -1/2 & -1/2 \\ -1/2 & 1 & 1/2 \\ -1/2 & 1/2 & 1 \end{pmatrix}.$$

Since A is $1/2$ times the Hessian matrix, A is positive definite (in particular positive semidefinite), and hence $f(x, y, z) \geq 0$ for all $x, y, z \in \mathbb{R}$ according to the definition in Section 3.7.2.

An alternative solution is to show that $(0, 0, 0)$ is a critical point for f with $f(0, 0, 0) = 0$. From Theorem 8.19, one can then conclude that $f(x, y, z) \geq 0$ for all $x, y, z \in \mathbb{R}$, or use Theorem 8.9(i) together with Theorem 6.12.

PROBLEM 3

- (a) The points $(0, 1), (1, 0), (1, 2), (2, 1) \in \mathbb{R}^2$ are assigned labels $+1, -1, -1, +1$ respectively:



Show that if the line $L = \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\}$ separates the points, then

$$\begin{aligned} b + c &> 0 \\ 2a + b + c &> 0 \\ a + c &< 0 \\ a + 2b + c &< 0. \end{aligned} \tag{3}$$

ANSWER:

If L separates the points, then according to Section 9.1 (see also 5.2.3), $ax + by + c > 0$ for all points (x, y) with label $+1$ and $ax + by + c < 0$ for all points (x, y) with label -1 . This gives the inequalities

$$\begin{aligned} b + c &> 0 \\ 2a + b + c &> 0 \\ a + c &< 0 \\ a + 2b + c &< 0 \end{aligned}$$

when substituting the points.

- (b) Show using Fourier-Motzkin elimination that the inequalities in (3) cannot be solved, and hence that there exists no line that separates the points.

ANSWER:

We are asked to explicitly use Fourier-Motzkin elimination cf. Section 4.5. We start by eliminating c :

$$\begin{aligned} -b &< c \\ -2a - b &< c \\ c &< -a \\ c &< -2b - a \end{aligned}$$

This gives the inequalities

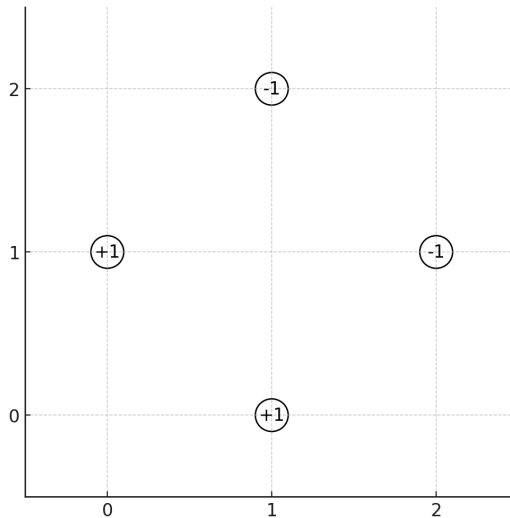
$$\begin{aligned} -b &< -a \\ -b &< -2b - a \\ -2a - b &< -a \\ -2a - b &< -2b - a \end{aligned}$$

Here the second inequality gives $-b - a > 0$, while the third inequality gives $-a - b < 0$. This is a contradiction. Alternatively, one can also continue by eliminating b :

$$\begin{aligned} a &< b \\ -a &< b \\ b &< -a \\ b &< a \end{aligned}$$

which gives the contradiction $a < a$.

(c) We now assign the same points $(0, 1), (1, 0), (1, 2), (2, 1)$ labels $+1, +1, -1, -1$ respectively:



Set up an optimization problem for finding the best separating line $L = \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\}$ that now separates the points. Show, without drawing (without geometric considerations), that $a = -1, b = -1$, and $c = 2$ is an optimal solution to the optimization problem.

ANSWER:

The optimization problem is set up according to Equation (9.8) in Section 9.1. Here $\alpha = (a, b)$ and $\beta = c$, and the optimization problem becomes

$$\begin{aligned} \min \quad & a^2 + b^2 \\ & b + c \geq 1 \\ & a + c \geq 1 \\ & 2a + b + c \leq -1 \\ & a + 2b + c \leq -1 \end{aligned}$$

Cf. (9.24), we set up the KKT conditions for this optimization problem:

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ \lambda_3 &\geq 0 \\ \lambda_4 &\geq 0 \\ -b - c + 1 &\leq 0 \\ -a - c + 1 &\leq 0 \\ 2a + b + c + 1 &\leq 0 \\ a + 2b + c + 1 &\leq 0 \\ \lambda_1(-b - c + 1) &= 0 \\ \lambda_2(-a - c + 1) &= 0 \\ \lambda_3(2a + b + c + 1) &= 0 \\ \lambda_4(a + 2b + c + 1) &= 0 \\ 2a - \lambda_2 + 2\lambda_3 + \lambda_4 &= 0 \\ 2b - \lambda_1 + \lambda_3 + 2\lambda_4 &= 0 \\ -\lambda_1 - \lambda_2 + \lambda_3 + \lambda_4 &= 0\end{aligned}$$

By substituting $a = -1$, $b = -1$, and $c = 2$ into the KKT conditions, we get the equations (the other conditions are satisfied):

$$\begin{aligned}-\lambda_2 + 2\lambda_3 + \lambda_4 &= 2, \\ -\lambda_1 + \lambda_3 + 2\lambda_4 &= 2, \\ -\lambda_1 - \lambda_2 + \lambda_3 + \lambda_4 &= 0.\end{aligned}$$

These equations are satisfied for

$$\begin{aligned}\lambda_2 &= 2 - \lambda_1, \\ \lambda_3 &= 2 - \lambda_1, \\ \lambda_4 &= \lambda_1.\end{aligned}$$

For $\lambda_1 \in [0, 2]$, we have $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \geq 0$, and hence the KKT conditions are satisfied and the solution is optimal by Theorem 9.30(ii).